Answer Key — Solubility of Gases

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1	**(b) oxygen** — dissolved oxygen sustains aquatic life.
Q2	**(b) decreases** — gas solubility decreases with temperature.
Q3	**(a) cold water** — cold water holds more oxygen.
Q4	**(b) uniform** — dissolved gas spreads evenly — a uniform mixture.
Q5	**(a) Both true, R explains A** — less oxygen in warm water harms fish — exactly as R says.
Q6	**decreases** — gas solubility falls with rising temperature.
Q7	**dissolved** — dissolved oxygen sustains aquatic life.

Short answer

Q8	Yes; e.g. oxygen dissolves in water (only a little), and this dissolved oxygen keeps aquatic life such as fish and plants alive.
Q9	Gas solubility decreases as temperature rises — the opposite of most solids, whose solubility increases with temperature.
Q10	Cold water holds more dissolved oxygen, giving fish and other aquatic organisms more oxygen to breathe.

Long / application

Q11	Hot water raises the river's temperature, and gas solubility decreases with temperature, so the warmer water holds less dissolved oxygen. With less oxygen available, fish and other aquatic organisms can suffocate, so thermal pollution harms aquatic life.
Q12	For most solids, solubility increases with temperature — hot tea dissolves more sugar. For gases, solubility decreases with temperature — a warm fizzy drink loses its gas and goes flat. So heating helps dissolve solids but drives gases out.

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